

Do Incentives to Review Help the Market? Evidence from a Field Experiment on Airbnb

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Abstract. Many online reputation systems operate by asking volunteers to write reviews for free. As a result, a large share of buyers do not review, and those who do review are self-selected. This can cause the reputation system to miss important information about seller quality. We study the extent to which a platform can improve market outcomes by attempting to increase the amount and quality of information collected by its reputation system. We do so by analyzing a randomized experiment conducted by Airbnb. In the treatment, buyers were offered a coupon to review listings that had no prior reviews. In the control, buyers were not offered any incentive to review. We find that, although the treatment induced additional reviews that were more negative on average, these reviews did not affect the number of nights sold or total revenue. Furthermore, we find that, contrary to the treatment's intended effect, Airbnb's incentivized program caused transaction quality for treated sellers to fall. We examine how the quality of the induced reviews, market conditions, and the design of Airbnb's reputation system can explain our findings.

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1. Introduction

Reputation systems are used by nearly every digital marketplace to help match buyers and sellers. Although existing reputation systems are considered critical to the success of digital marketplaces, they suffer from a variety of imperfections documented in previous research (Tadelis 2016). A major imperfection present in many reputation systems is that, because online reputation is a public good,¹ not all buyers leave reviews, and the subset who do are self-selected.

In theory, increasing review rates can improve market outcomes for at least two reasons (Avery et al. 1999). First, the more often reviews are left by buyers, the faster other buyers learn about sellers' quality. Acemoglu et al. (2022) show that increasing the speed of learning about seller quality increases welfare by facilitating better matches. Second, when the reviews collected by a reputation system are nonrepresentative, buyers may form biased beliefs about the quality of sellers, leading to the

formation of bad matches. For instance, if reviews are more likely to be submitted after a positive experience than a negative one, future buyers may overestimate the quality of sellers (Dellarocas and Wood 2007, Nosko and Tadelis 2015). In concordance with this theory, platforms such as Amazon, Wayfair, and Walmart use incentives to improve review rates.² However, the effectiveness of these incentivized review programs in helping the market is unknown.

Using a large-scale and costly experiment conducted on Airbnb, we study whether incentivizing reviews improves market outcomes and find that it does not.³ In the treatment group, guests to treated listings were sent an email offering them a \$25 Airbnb coupon in exchange for a review if they had not reviewed within (typically) eight or nine days after checkout, whereas guests to control listings were instead sent an email reminding them to leave a review.

The incentivized review policy increased the review rate by 53% relative to the review rate in the control

group and induced reviews with lower star ratings on average. These results confirm prior work that shows that incentives and nudges are effective at inducing additional reviews (Burtch et al. 2018, Karaman 2020, Marinescu et al. 2021) and that reviews induced by incentives are more likely to contain information about lower quality transactions (Burtch et al. 2018, Marinescu et al. 2021). We also find that, compared to nonincentivized reviews with the same star rating, incentivized reviews corresponded to worse guest experiences. In other words, although incentivized reviews had a lower average rating, they were more inflated at a given star rating than nonincentivized reviews.

Although the incentivized review program impacted the number and type of reviews left by guests, it had neutral-to-negative impacts on the business outcomes of the platform and its sellers. Incentivized reviews did not affect the total quantity (nights) sold for treated listings, but did cause a change in the composition of transactions. Specifically, treated listings had more transactions, but stays at treated listings lasted fewer nights on average. As a consequence, the revenue effects of incentivized reviews are not statistically distinguishable from zero. Furthermore, contrary to the policy's intent, incentivized reviews failed to improve transaction quality and, according to some measures, resulted in worse matches. In particular, the treatment did not affect complaint rates and reduced the posttransaction usage of Airbnb by guests staying with listings after the review. We argue that this is, in part, related to our finding that incentivized reviews were less correlated with true transaction quality.

Our inability to detect any positive seller- or market-level effects stemming from the introduction of the incentivized review policy can be attributed to both market structure and the design of Airbnb's reputation system. On the market structure front, if additional reviews arrive quickly after experimental assignment, then the effect of receiving an incentivized review (or not) is muted. Alternatively, if sellers cannot obtain additional transactions and reviews unless they receive an initial review, then the effect of incentivized reviews is large. We show that reviews from other transactions arrived quickly for listings in our experimental sample. Furthermore, listings on Airbnb have capacity constraints that limit the extent to which any treatment can increase quantity sold. Although our empirical results are specific to Airbnb, fast review arrival rates and capacity constraints are common to many large online marketplaces.⁴

Incentivized reviews may have also had small effects on listing outcomes because of Airbnb's reputation system design at the time of the experiment (2014–2016). In particular, star ratings (as opposed to the text and number of reviews) were rounded to the nearest half star and were only displayed once a listing had at least three

ratings. As a result, ratings induced by the incentivized review program were averaged with at least two other ratings when displayed to guests on Airbnb. This averaging, rounding, and censoring attenuated perceived differences between the ratings of control and treatment listings. We expect similar mechanisms to exist in other platforms that round to half stars, including Amazon, Etsy, and Yelp.

An important implication of our findings is that institutional details, such as market conditions and reputation system design, are critical for understanding the value of reviews and the effects of reputation system-related treatment interventions. We do not claim that reviews and reputation systems have little value. In fact, prior work shows that reputation systems substantially increase consumer surplus (Wu et al. 2015, Lewis and Zervas 2016, Reimers and Waldfogel 2021). Instead, we show that additional reviews do not help when listings are expected to receive a flow of reviews and when review ratings are displayed as rounded averages.

2. Literature Review

We contribute to several related research themes within the study of online reputation systems. The first and most directly relevant of these research themes focuses on incentivized review policies. In an early contribution to the literature studying online reviews, Avery et al. (1999) show that, in equilibrium, evaluations are underprovided relative to what is optimal from a welfare perspective unless would-be reviewers are offered an incentive to leave feedback. Building on this insight, a number of recent papers study the effectiveness of incentivized review policies and nudges to review at increasing review rates in online settings. Burtch et al. (2018), Marinescu et al. (2021), and Karaman (2020) experimentally document that incentives and solicitations offered by platforms can not only increase review rates, but also generate more representative reviews.⁵ In an adjacent stream of work, Li (2010), Li and Xiao (2014), Cabral and Li (2015), and Li et al. (2020) study policies in which sellers (rather than the platform) offer rebates for reviews.

In a piece of research that is particularly closely related to our work, Pallais (2014) experimentally measures the effects of an intervention in which new sellers in an online labor market are both hired and reviewed. She finds that hiring workers and leaving positive feedback has large positive effects on subsequent demand. In contrast, the policy we study generates reviews only for the subset of sellers who are able to transact before receiving their first review. An advantage of this policy relative to the one studied in Pallais (2014) is that it is less expensive for platforms to implement because they do not have to pay to hire new sellers. However, a disadvantage is that the incentivized first review policy does not solve the “cold-start” problem for listings that are unable to transact prior to receiving their first review.

Beyond work that studies incentivized review policies and nudges to review, there is a large body of work that studies bias in reputation systems. Several papers show that nonincentivized reviews exhibit a positivity bias (Dellarocas and Wood 2007, Nosko and Tadelis 2015, Fradkin et al. 2021, Brandes et al. 2022, Filippas et al. 2022). This positivity bias has a number of causes, including a higher propensity to post feedback when satisfied (Dellarocas and Wood 2007), reciprocity and retaliation (Dellarocas and Wood 2007, Fradkin et al. 2021), selection effects that dictate which buyers transact with which sellers (Nosko and Tadelis 2015), “reputation inflation” (Filippas et al. 2022), and differential attrition of those with moderate experiences (Brandes et al. 2022).⁶ Our results highlight that positivity bias in incentivized ratings may even be worse than in nonincentivized ratings. In particular, conditional on a given star rating, reviews induced by the incentive tended to correspond to worse experiences.

There is also an emerging research literature that aims to model the dynamics with which consumers learn about seller quality under different ratings systems (Besbes and Scarsini 2018, Papanastasiou et al. 2018, Ifrach et al. 2019, Acemoglu et al. 2022). Our work is closely related to Besbes and Scarsini (2018) and Acemoglu et al. (2022), which both compare consumer learning dynamics under different reputation system designs. We argue in this paper that one of the reasons that the additional reviews induced by Airbnb’s incentivized first review program failed to have a measurable impact on market outcomes is the design of Airbnb’s review system, which has characteristics of the “summary statistics only” systems discussed in both Acemoglu et al. (2022) and Besbes and Scarsini (2018).⁷

We contribute to these research literatures by experimentally studying the implications of a change to the design of Airbnb’s reputation system not only on the number and type of reviews left by guests, but also on subsequent market outcomes.⁸ Although one might expect an incentivized first review program such as the one we study to generate additional information on seller quality that yields more and better matches on the platform, we do not find this to be the case empirically; we instead find that the program had no effect on demand and negative effects on match quality. We argue that this is due to market conditions in the markets in which most of our sample listings were located, the manner in which Airbnb’s reputation system aggregates and displays review information, and the capacity constraints of Airbnb listings.

3. Setting and Experimental Design

We analyze an experiment conducted on Airbnb, the largest online marketplace for peer-to-peer short-term accommodations, from April 12, 2014, to May 17, 2016.

At the time of the experiment, Airbnb’s review system worked as follows. After the guest’s checkout, both the host and the guest were asked via email, web notifications, and app notifications to review each other. Both guest and host reviews consisted of both numeric and textual information. The text of reviews written by guests was displayed on listing pages in reverse chronological order. The numeric overall and category ratings, which were on a one- to five-star scale, were displayed as averages across all transactions rounded to the nearest half star. Rounded average ratings were only visible on listing pages once a listing had received at least three reviews; before that, only review text was visible on a listing page. The number of reviews a listing received was visible on both the search and listing pages as long as the listing had one review, meaning that reviews were able to have effects through both the search and listing pages.

Prior to July 2014, guest and host reviews were visible to both the counterparty and the public immediately after submission, and reviews needed to be submitted within 30 days of checkout. Beginning in July 2014, a simultaneous reveal review system was put in place (Fradkin et al. 2021). Under the simultaneous reveal system, guests and hosts had 14 days after checkout to submit a review, and reviews were only publicly displayed after both parties submitted a review or 14 days had elapsed. Because our experiment ran from April 2014 to May 2016, the vast majority of our data was collected under the simultaneous reveal review system.

Experiment randomization was conducted at the Airbnb listing level. In order to be eligible for enrollment in the experiment, a listing needed to meet the following criteria:

- It needed to have been booked.
- It needed to have no prior reviews.
- Following a guest checkout, the guest must not

have reviewed within a threshold number of days. This threshold was typically eight or nine days throughout most of our sample with the specific number of days varying because of idiosyncratic details of Airbnb’s email dispatching system.⁹

Across Airbnb’s entire platform, guests who had not reviewed within the threshold number of days received an email reminding them to review. For stays at control listings that met the criteria, guests received the standard review reminder email. For stays at treatment listings that met the criteria, guests received a reminder email that also offered a \$25 Airbnb coupon in exchange for leaving a review. These coupons expired one year after being issued and needed to be used on stays with a minimum cost of \$75. Online Appendix Figure B.1 shows the email sent to guests who stayed at treatment listings without reviews during the experiment. In our sample, 326,602 listings were assigned to the control and 328,266 listings were assigned to the treatment. The

experiment used a well-tested system at Airbnb for randomization into treatment conditions and achieved good balance on pretreatment covariates (Online Figure C.4).

Because randomization was conducted at the listing level, many of our analyses utilize the concept of a listing's focal stay, which is the first transaction for which a listing is in either the treatment or control. This is in contrast to subsequent stays that also may result in reviews, but may have been affected by the presence of a review for the focal stay.

4. Effects of Experiment on Reviews

Incentivized reviews can only have effects on market outcomes if they have an effect on the quantity and types of reviews that are submitted by guests. Thus, we first measure the effects of incentives on these review-related outcomes. We find that the treatment induced additional reviews for focal stays and those reviews tended to have lower ratings and more negative text sentiment on average.

Table 1, panel A, shows the effect of the treatment on review rates and the distribution of numerical ratings before conditioning on a review being left. The first thing that is apparent is that the treatment was effective at increasing the rate at which guests submitted reviews: the treatment increased the review rate by 12.86 percentage points (pp) from 24.19% to 37.05%. Because of this increase in review rate, before conditioning on a review being left, the treatment also increased the number of

five-star reviews (6.28 pp), four-star reviews (4.64 pp), three-star reviews (1.44 pp), two-star reviews (0.31 pp), and one-star reviews (0.2 pp).

In addition to increasing the absolute number of reviews left at each star rating, the treatment also changed the relative frequency of ratings. Table 1, panel B, shows that, conditional on a review, ratings of treated listings were, in fact, lower than ratings of control listings; the treatment caused the average rating left by guests to drop by 0.07 stars from 4.48 to 4.41. This effect was driven by a decrease in the rate of five-star reviews and an increase in the rate of two- to four-star reviews. In other words, although the treatment led to an across-the-board increase in the number of reviews at all star ratings, the increase was larger for lower than higher ratings.

The reviews induced by the incentivized review treatment not only had lower star ratings; they also had more negative text. After classifying the sentiment of the review text for each review in our sample, we find that the percentage of review text classified as positive decreased by 0.8% in the treatment group.¹⁰ This difference in text sentiment disappears once we condition on the review's star rating, suggesting that the effects we report on star ratings and review text are consistent with each other.

The differences in ratings that we observe should correspond to differences in the observable and/or unobservable characteristics of reviewed transactions across the experiment's treatment conditions. Table 2 reports the effect of the incentivized review treatment on the

Table 1. Differences in Ratings by Treatment

Panel A. All trips						
	No review (1)	One star (2)	Two stars (3)	Three stars (4)	Four stars (5)	Five stars (6)
Constant	0.7581*** (0.0007)	0.0042*** (0.0001)	0.0045*** (0.0001)	0.0140*** (0.0002)	0.0671*** (0.0004)	0.1521*** (0.0006)
Incentivized review	−0.1286*** (0.0011)	0.0020*** (0.0002)	0.0031*** (0.0002)	0.0144*** (0.0004)	0.0464*** (0.0007)	0.0628*** (0.0010)
R ²	0.01946	0.00019	0.00040	0.00250	0.00655	0.00657
Observations	654,868	654,868	654,868	654,868	654,868	654,868
Panel B. Reviewed trips						
	One star (1)	Two stars (2)	Three stars (3)	Four stars (4)	Five stars (5)	Positive text (6)
Constant	0.0175*** (0.0005)	0.0186*** (0.0005)	0.0578*** (0.0008)	0.2773*** (0.0016)	0.6288*** (0.0017)	0.9405*** (0.0010)
Incentivized review	−0.0008 (0.0006)	0.0019** (0.0006)	0.0188*** (0.0011)	0.0290*** (0.0021)	−0.0489*** (0.0022)	−0.0080*** (0.0013)
R ²	8.93 × 10 ^{−6}	4.25 × 10 ^{−5}	0.00131	0.00097	0.00238	0.00026
Observations	200,611	200,611	200,611	200,611	200,611	135,670

Notes. The table displays the results of regressions in which the outcome is a rating outcome and the only covariate is the treatment indicator. Panel A displays the regression for all trips, whereas panel B displays results for just reviewed trips. Note that panel B, column (6), has fewer observations because reviews in a foreign language could not be classified by the sentiment analyzer. Robust standard errors are displayed in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 2. Differences in Characteristics of Reviewed Transactions Treatment vs. Control

	Pro. host (1)	Revenue (2)	Nights (3)	Price (nightly) (4)	Bedrooms (5)	Complaint (6)	GFSR (7)
Constant	0.4499*** (0.0018)	639.3*** (3.681)	6.738*** (0.0400)	116.0*** (0.5065)	1.602*** (0.0042)	0.0128*** (0.0004)	0.6894*** (0.0021)
Incentivized	0.0187*** (0.0023)	−33.52*** (4.730)	−0.3274*** (0.0514)	−1.308* (0.6507)	−0.0307*** (0.0053)	−0.0002 (0.0005)	−0.0031 (0.0027)
R ²	0.00034	0.00025	0.00020	2.02×10^{-5}	0.00016	7.8×10^{-7}	1.56×10^{-5}
Observations	199,654	199,654	199,654	199,654	199,654	199,654	82,182

Notes. The table displays the results of regressions in which the outcome is a trip characteristic and the regressor is whether the review was incentivized (versus control). “Pro. host” is a host with more than one listing, “Complaint” takes the value of one if the customer had a complaint to Airbnb during the trip, and “GFSR” is the guest’s rate of giving five-star ratings for any prior trips the guest has taken (note that this has fewer observations because many guests had not previously had an Airbnb trip). Robust standard errors are displayed in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

prevalence of different trip- and user-level characteristics among reviewed focal transactions. We find that reviews in the treatment group were more likely to come from trips to lower capacity Airbnb listings (column (5)) that were run by multilisting hosts (column (1)). These trips were also on average lower value (column (2)) because of both shorter duration (column (3)) and lower price per night (column (4)). Overall, these effects are consistent with the notion that the subset of guests who were responsive to a \$25 coupon incentive were more likely to be budget-conscious travelers.

The lower average ratings we observe in induced reviews could also be driven, as posited by much of the literature, by lower quality transactions. We can’t directly observe transaction quality, but we do have a sparse proxy for very bad transactions: the customer complaint rate. In column (6), we show that customer complaint rates did not differ in a statistically significant manner between treatment and control transactions. Note that this lack of difference does not preclude differences in transaction quality that may not be captured by customer complaints.

One other reason that incentivized reviews may have different ratings than nonincentivized reviews is that the guests who review because of incentives may have had different reviewing styles. For example, these guests may tend to judge Airbnb listings more harshly than other reviewers. We test for this possibility by measuring the reviewing behavior of guests prior to the focal transaction. Specifically, we compare the average historical guest-level five-star review rate (GFSR) for reviewed trips in the control and treatment groups. We find that the observed difference in review ratings and text is not explained by guest harshness (column (7)).

Although the incentivized review treatment reduced the amount of differential nonresponse in reviewing, it did not eliminate it entirely. This is apparent in Online Figure C.6, which compares the trip- and user-level characteristics of reviewed transactions in the treatment group to those of nonreviewed transactions in the treatment group. First, reviewed transactions in the

treatment were less likely to have complaints than nonreviewed transactions in the treatment, implying that, even under the incentivized review policy, many extremely low-quality transactions were not reviewed. Second, reviewed transactions in the treatment had lower value, duration, and price per night than nonreviewed transactions in the treatment, further supporting the notion that less budget-conscious guests are not as responsive to the \$25 Airbnb coupon incentive.

In summary, the incentivized review treatment induced additional reviews, and those reviews were on average more negative. These reviews tended to come from guests who were more price-sensitive and, thus, more responsive to the offer of a \$25 Airbnb coupon. Although the provision of these reviews reduced the severity of the selection bias in which guests leave feedback, there is evidence that, even in the treatment, many guests who were less price-sensitive and/or who experienced low-quality transactions still did not review.

5. Effects of Incentivized Reviews on Market Outcomes

We’ve shown that the treatment induced reviews, which changed the information set of subsequent buyers. It is, however, theoretically ambiguous if and how these reviews affected market outcomes.¹¹ In this section, we measure the extent to which the incentivized review program changed outcomes such as purchase rates, platform revenues, and match quality.

5.1. The Effect of Incentivized Reviews on Demand for a Listing

We first measure the effect of the treatment on a listing’s subsequent demand. Although prior literature shows that positive (negative) reviews increase (decrease) demand, the direction and magnitude of the effect of our treatment on demand is ambiguous. This is both because the review incentive induced a mixture of positive and negative reviews and because the extent to which an individual positive or negative review impacts demand in

our setting is unknown. To measure the effects of the treatment on demand, we estimate linear regressions of the following form:

$$y_l = \beta_0 + \beta_1 T_l + \epsilon_l, \quad (1)$$

where T_l is an indicator for whether the listing, l , had a guest who was sent a treatment email offering a coupon in exchange for the review and y_l is one of the following proxies for guest demand in the 120 days immediately following the focal trip checkout: the total number of listing views, the total number of transactions, the total number of nights across all transactions, and the total booking value across all transactions.

Table 3, panel A, shows that views (column (1)) and reservations (column (2)) increased. On the other hand, the total nights of stay (column (3)) and booking value (column (4)) exhibit small effects that are statistically indistinguishable from zero. Both hosts and the platform care most about total nights booked and the revenue from transactions, meaning that, according to the most meaningful metrics, the treatment did not increase demand on average.¹²

A key component of any incentivized review program is imperfect compliance, meaning that not everyone responds to an incentive by reviewing. If compliance in our experiment were sufficiently low, then we may not find any effect of a review simply because too few additional reviews were produced. To better understand what our experiment implies about the effect of a review, we use a two-stage least squares estimator. In particular, we measure the local average treatment effect of an incentivized review using the following second

stage equation:

$$y_l = \beta_0 + \beta_1 R_l + \epsilon_l, \quad (2)$$

where R_l takes the value of one if the listing, l , was reviewed for the focal transaction in the experiment and the instrument is the treatment assignment in the incentivized review experiment.¹³

Table 3, panel B, displays our estimates of the local average treatment effect of an incentivized review. We find that incentivized reviews generated more attention and transactions for listings. Specifically, the treatment led to a 7.1% increase in views and a 9.1% increase in transactions, which translates to an additional 0.326 transactions per listing, in the 120 days immediately following the focal checkout. The fact that the treatment increased listing views and transactions by similar percentages suggests that the effect of an incentivized review on transactions was primarily driven by an increase in clicks from the search page to the listing page. In Online Appendix B.4, we show that this change in clickthrough rate was because the number of reviews a listing had was displayed on the search page rather than changes in listings' algorithmic search ranking. We also find that the increase in transactions because of an incentivized review did not translate to an increase in the number of nights booked, and the 95% confidence interval around our point estimate rules out effects of an incentivized review on nights that are larger than 8%. We view this as evidence against large effects of incentivized reviews on demand.

Given that the number of transactions rose but the number of nights did not, it must be the case that

Table 3. Effects on Listing Outcomes (120-Day Horizon)

Panel A. Intent to treat				
	Views (1)	Reservations (2)	Total nights (3)	Booking value (4)
Constant	753.3*** (2.317)	3.665*** (0.0121)	15.04*** (0.0469)	1,638.6*** (6.525)
Assigned to treatment	6.725* (3.359)	0.0416* (0.0171)	0.0227 (0.0661)	4.262 (9.202)
Observations	654,595	654,595	654,595	654,595
Panel B. Local average treatment effect				
	Views (1)	Reservations (2)	Total nights (3)	Booking value (4)
Constant	740.6*** (8.118)	3.586*** (0.0418)	14.99*** (0.1616)	1,630.6*** (22.48)
Reviewed	52.70* (26.32)	0.3259* (0.1344)	0.1782 (0.5180)	33.40 (72.10)
Observations	654,595	654,595	654,595	654,595

Notes. The table displays the results of regressions in which outcomes are measured between the checkout of the focal trip and 120 days afterward. Panel A displays the intent to treat estimates, whereas panel B displays local average treatment effect estimates from a two-stage least squares regression in which the first stage is a regression of whether a focal trip was reviewed on the treatment assignment. Robust standard errors are displayed in parentheses.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Table 4. Characteristics of Subsequent Trips

Panel A. Intent to treat				
	Nights per trip (1)	Trip revenue (2)	Price per night (3)	Lead time (days) (4)
(Intercept)	4.207*** (0.0098)	396.5*** (1.340)	103.7*** (0.3128)	17.29*** (0.0384)
Assigned to treatment	−0.0403** (0.0136)	−3.473 (1.882)	−0.4688 (0.4403)	0.0272 (0.0543)
R ²	7.84 × 10 ^{−6}	7.37 × 10 ^{−6}	4.68 × 10 ^{−6}	4.79 × 10 ^{−7}
Observations	2,389,288	2,389,288	2,389,288	1,892,755
Panel B. Local average treatment effect				
	Nights per trip (1)	Trip revenue (2)	Price per night (3)	Lead time (days) (4)
(Intercept)	4.290*** (0.0358)	403.7*** (4.929)	104.7*** (1.151)	17.23*** (0.1394)
Reviewed	−0.3133** (0.1059)	−26.99 (14.64)	−3.643 (3.422)	0.2088 (0.4161)
R ²	−0.00041	−0.00022	4.68 × 10 ^{−5}	0.00013
Observations	2,389,288	2,389,288	2,389,288	1,892,755

Notes. This table displays regressions at a transaction level of transaction characteristics on the treatment. All transactions for listings in the experiment that occur within 120 days of the checkout of the focal stay are considered. The regression for lead time includes fewer observations because we considered only trips for which the check-in occurred within 120 days. Panel A displays intent to treat effects, whereas panel B displays local average treatment effects from a two-stage least squares regression. Standard errors are clustered at the listing level.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

incentivized reviews changed the types of trips that occur. We investigate this by estimating trip-level regressions on a data set consisting of all transactions that were booked within 120 days of assignment with standard errors clustered at the listing level. Table 4 shows the results of these regressions. We find that there are no statistically significant differences with respect to most trip characteristics with the exception of nights per trip, which fell by 1%. Put differently, although the treatment increased the number of bookings that occurred, the average booking was shorter in duration such that there was not a statistically significant treatment effect on total nights or booking value.

To summarize, the net effect of Airbnb's incentivized review program on quantities sold and revenue was statistically indistinguishable from zero despite treated listings receiving more listing views and transactions because of the incentivized review program. This is because, although the receipt of an incentivized review led to a greater number of transactions, these transactions were shorter on average.

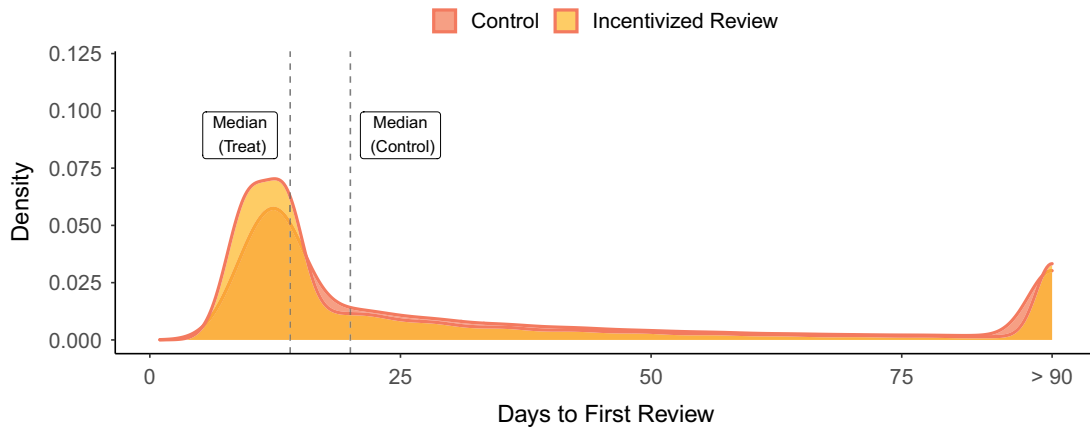
5.2. Why Don't Incentivized Reviews Affect Demand?

There are several reasons why the incentivized review treatment failed to increase demand in the 120 days following treatment assignment despite increasing the number of booking listings received.¹⁴ The first of these is the fact that listings in our sample were typically able to generate transactions even without a review. Thus, trips occurring after the focal transaction

provided additional opportunities for listings to receive their first review. Reviews from nonfocal trips often arrived quickly and attenuated differences in the distributions of first ratings for treatment and control listings.

More specifically, because the review incentive was only offered to guests after checkout, every listing in the experiment was, by definition, able to receive at least one booking without having any reviews. This means that, at least for some guests, the presence of a first review was not pivotal in their choice of listing. One reason that guests take a chance with a nonreviewed listing is that many Airbnb markets are supply constrained.¹⁵ As a result, guests are shown listings without reviews and sometimes book these listings.

The focal trips on which we focus on in our analysis are not anomalies; 45% of listings in the experiment had more than one booking prior to the checkout of the focal trip, not to mention bookings occurring after the focal trip ended. Each of these additional bookings offered an opportunity for the listing to receive a review, and these opportunities added up. By August 2016, 72.8% of listings in the control group and 78.4% of listings in the treatment group had received at least one review. This 5.6% difference is less than half as large as the effect of the treatment on the review rate for the focal transaction (13%). Furthermore, not much time elapsed during which treatment listings had a review and control listings did not because, even in the control group, reviews arrived quickly: Figure 1 shows that the difference between the median time to first review in the control and treatment groups is only six days.

Figure 1. (Color online) Distribution of Days to First Review

Notes. The figure plots the distribution of the time of arrival for the first review for treated and control listings. The time is calculated as the difference in days between the date of the arrival of the first review and the checkout of the transaction for which the experimental assignment occurred.

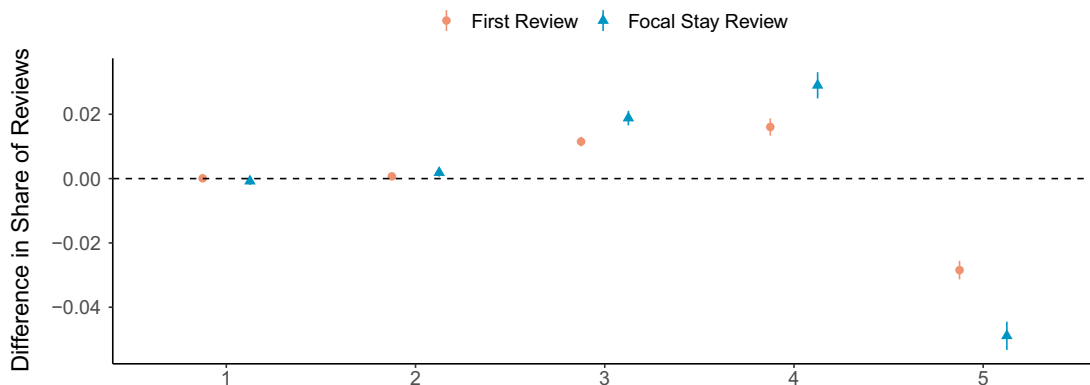
Even though the effect of the treatment on the percentage of listings receiving at least one review shrinks after taking into account reviews from nonfocal transactions, one might still expect differences in the control and treatment first rating distributions to result in demand effects. However, Figure 2 shows that, after taking into account reviews from additional bookings, the differences between the first ratings distributions of treatment and control reviews (previously reported in Section 4) are also attenuated. This is likely because of a number of factors, including the fact that reviews from nonfocal trips in the treatment consist of a mix of nonincentivized reviews (left within eight to nine days) and incentivized reviews. Regardless of the reasons behind this attenuation, a smaller difference in the distributions of first ratings that listings received also reduces the likelihood that we observe an effect of the treatment on demand.

A second reason that incentivized reviews had no effect on demand is the manner in which Airbnb displayed reviews at the time of the experiment. Ratings were not

shown for every review, but were instead averaged and rounded to the nearest half star.¹⁶ Furthermore, whereas review text was publicly visible after a listing received its first review, average ratings were only shown after a listing had received three reviews. Because numerical rating information was not available to some guests when they viewed the listing page and, for guests who observed numerical information, any differences had been attenuated by at least two other reviews, the likelihood that one review would change a guest's perception of the listing is low.

A third reason is that the capacity of Airbnb listings is limited. In particular, unlike markets for mass-produced goods, only one buyer can book an Airbnb listing per night. As a result, increased listing views caused by the treatment could only increase quantity sold if those listing views resulted in bookings for marginal nights.

In theory, it's also possible that demand was not affected by incentivized reviews because sellers updated their nightly prices after receiving a review. However,

Figure 2. (Color online) Effect on Review Ratings (Conditional on Review)

Notes. The figure plots the estimate and 95% confidence interval for differences in the share of reviews with each rating type between treatment and control listings. “Focal stay review” refers to any review that occurred for the first transaction for a listing that was eligible for the experimental treatment. “First review” refers to the first review ever received by a listing. Note that, to be eligible for the experiment, the listing must have had no prior reviews.

Table 5. Effects of Treatment on Transaction Quality

	Complaint	Reviewed	Star rating	Guest nights	
	(1)	(2)	(3)	(4)	(5)
Constant	0.0101*** (0.0001)	0.6475*** (0.0006)	4.529*** (0.0014)	5.591*** (0.0217)	
Treatment	-6.52×10^{-5} (0.0001)	0.0048*** (0.0009)	-0.0060^{**} (0.0020)	-0.0766^{**} (0.0296)	-0.0548^* (0.0245)
R ²	1.06×10^{-7}	2.52×10^{-5}	1.53×10^{-5}	6.48×10^{-6}	0.20805
Observations	2,431,085	2,431,085	1,579,132	2,431,085	2,431,085
Controls	No	No	No	No	Yes
Guest region fixed effects					✓
Checkout week fixed effects					✓
Number of nights fixed effects					✓
Number of guests fixed effects					✓

Notes. This table displays regressions measuring the effects of the treatment (the guest receiving an email with an offer of a coupon in exchange for a review) on measures of transaction quality. The set of transactions considered for this regression includes all transactions for which the checkout date was between the checkout date of the focal transaction and 360 days after. “Complaint” refers to whether a guest submitted a customer service complaint to Airbnb, “Reviewed” refers to whether the guest submitted a review, “Star rating” refers to the star rating of any submitted reviews, “Guest nights” refers to the number of transacted nights for a guest in the 360 days postcheckout. Control variables in column (5) include the log of transaction amount, the number of times the guest has reviewed and reviewed with a five-star rating in the past, the prior nights of the guest, whether the guest has an “about” description, and guest age on the platform.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

our results suggest that Airbnb hosts did not do so.¹⁷ This behavior is consistent with Huang (2022), who finds that sellers on Airbnb are often inattentive or constrained in changing prices when responding to demand fluctuations.

To summarize, a number of explanations contribute to the fact that the incentivized review program we study failed to impact overall demand despite the fact that the treatment affected the number and quality of reviews left for focal transactions. First, in many cases, even listings assigned to the control eventually obtained a first review. These reviews arrived quickly after experimental assignment and attenuated the differences in the distribution of first ratings received by listings in our experiment. Furthermore, because numerical ratings were averaged and not visible until a listing received three reviews, it was difficult for one rating to appreciably change guests’ perceptions of a listing. Finally, because Airbnb listing nights are not mass-produced goods, demand effects were only possible if the treatment could cause bookings for marginal nights.

5.3. The Effect of Incentivized Reviews on Transaction Quality

Although the incentivized review program failed to yield a statistically significant impact on overall demand, it is possible the program affected the quality of matches that occurred on Airbnb. This could be the case if, for instance, the presence of a review or the review text changed the decision of guests about which listing to pick. To test for this possibility, we construct transaction-level customer satisfaction proxies and compare the average level of each of these proxies for posttreatment transactions to control and treatment listings. More concretely, for each

listing, l , we consider all transactions that occurred within 360 days of the focal stay checkout, did not have a cancellation, and had an observed payment. For this sample of transactions, we measure three customer satisfaction proxies: customer complaints, reviews, and customer return rates. Customer return rates are measured by the number of subsequent nights on the platform for guests staying at the listing posttreatment.¹⁸

Table 5 displays our results. Depending on the customer satisfaction proxy used, we find that the treatment had a neutral to negative effect on subsequent match quality. More specifically, we do not find a statistically significant impact of the treatment on the customer complaint rate (column (1)); however, this may stem from low statistical power because customer complaints are extremely rare on the platform (approximately 1% of transactions result in a complaint). Although the treatment had a positive impact on the review rate (column (2)), this was caused at least in part by the fact that, until treated listings received their first review, guests staying at those listings continued to receive the review incentive offer if they had not left a review within eight to nine days of checkout. Column (3) shows that, conditional on a subsequent trip being reviewed, the rating was worse in the treatment group. This is consistent not only with the idea that subsequent match quality was worse in the treatment group, but also with the fact that, in the treatment group, subsequent trips may have resulted in incentivized reviews if the listing did not receive a review for the focal transaction (recall that incentivized reviews have lower star ratings on average).¹⁹

Although these analyses suggest that the incentivized review treatment caused subsequent transaction quality to decrease, they rely on customer satisfaction

proxies, such as guest ratings, which may be subject to a number of reporting biases. Using subsequent behavior as a proxy for customer satisfaction provides a stronger result that is more straightforward to interpret. Column (4) of Table 5 shows that guests to treated listings stayed for 1.6% fewer posttransaction nights compared with guests to control listings. This effect remains statistically significant, albeit smaller in magnitude, after including controls for guest and trip characteristics (column (5)). The robustness of this result to the inclusion of these controls suggests that the apparent reduction in guests' subsequent platform usage is not because incentivized reviews induced matches with guests who had a lower baseline propensity to return to Airbnb. Instead, the simplest explanation for the results in columns (4) and (5) is that incentivized reviews induced worse matches and caused guests to use Airbnb less as a result. This could be the case if, for instance, incentivized five-star reviews were induced for low-quality listings, causing these listings to appear to subsequent guests as higher quality than they actually were.²⁰

We investigate whether this was, in fact, the case in our setting by considering how the predictive power of focal transaction ratings differs between the treatment

and control groups.²¹ If incentivized reviews were more inflated than nonincentivized reviews, then a five-star incentivized review should be less predictive of subsequent five-star nonincentivized reviews. If, on the other hand, incentivized reviews were less inflated than organic reviews, then incentivized reviews should be more predictive of subsequent five-star ratings. Table 6 displays the results of a regression in which the outcome is the rating of a subsequent review and the explanatory variable is the star rating of the focal review interacted with the treatment. We see that treatment five-star ratings were less associated with future five-star ratings than control five-star ratings regardless of the sample used (the next review (column (1)) or all subsequent reviews (column (2))) and whether covariates are added (columns (3) and (4)). This result suggests that five-star ratings in the treatment were indeed more inflated than five-star ratings in the control.

In summary, although we find that the incentivized review program affected the number and type of reviews left by Airbnb guests, we do not find evidence that the program affected quantities sold or revenues for listings or the platform. We identify a number of reasons that this is the case, including market conditions

Table 6. Predictive Power of Ratings: Treatment vs Control

	Rating			
	(1)	(2)	(3)	(4)
Constant	4.093*** (0.0509)	4.285*** (0.0272)		
Two stars	0.1036 (0.0640)	0.0347 (0.0367)	0.0984 (0.0635)	0.0276 (0.0351)
Three stars	0.2025*** (0.0540)	0.0932** (0.0297)	0.2097*** (0.0538)	0.0991*** (0.0285)
Four stars	0.3924*** (0.0513)	0.2460*** (0.0276)	0.4000*** (0.0511)	0.2484*** (0.0266)
Five stars	0.5625*** (0.0510)	0.3923*** (0.0273)	0.5549*** (0.0508)	0.3795*** (0.0263)
One stars × Treatment	0.0691 (0.0636)	−0.0101 (0.0383)	0.0729 (0.0631)	−0.0063 (0.0370)
Two stars × Treatment	−0.0680 (0.0496)	−0.0752* (0.0330)	−0.0616 (0.0490)	−0.0649* (0.0315)
Three stars × Treatment	−0.0316 (0.0222)	−0.0271 (0.0144)	−0.0252 (0.0220)	−0.0222 (0.0138)
Four stars × Treatment	−0.0295*** (0.0082)	−0.0322*** (0.0054)	−0.0295*** (0.0081)	−0.0298*** (0.0051)
Five stars × Treatment	−0.0199*** (0.0046)	−0.0183*** (0.0029)	−0.0196*** (0.0046)	−0.0172*** (0.0027)
Subsample	Next review	All subsequent reviews	Next review	All subsequent reviews
Covariates	No	No	Yes	Yes
Observations	134,245	1,308,783	134,089	1,304,491
R ²	0.02917	0.02041	0.05705	0.05192

Notes. This table displays a regression in which the star rating for stays after the focal stay is regressed on ratings for the focal stays and the treatment. Columns (1) and (3) use the sample of only the review following the review from the focal stay. Columns (2) and (4) use all subsequent reviews in the sample. Columns (3) and (4) include covariates for characteristics of subsequent stays. These are the log of transaction amount, the number of times the guest has reviewed and reviewed with a five-star rating in the past, the prior nights of the guest, whether the guest has an “about” description, guest age on the platform, guest region, checkout week, nights booked, and number of guests. Standard errors are clustered at the listing level in specifications (2) and (4).

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

on Airbnb, the design of Airbnb's reputation system, and capacity constraints for listings on the platform. Although we do not find effects of the treatment on overall demand, we do find that the treatment affected the quality of matches occurring on the platform. Specifically, match quality, as measured by both the star ratings that subsequent guests left and the rate at which guests returned to the platform, decreased. This is at least in part because ratings left under the incentivized review treatment were more inflated than those left under the status quo.

6. Discussion

In this paper, we analyze an experiment on Airbnb to determine the extent to which an incentivized review policy can improve market outcomes, such as demand and match quality. Although we find the incentivized review program was effective at inducing reviews and these reviews were more negative on average, the treatment did not affect overall demand and actually decreased the quality of subsequent matches on the platform.

The fact that we did not find an impact of the treatment on overall demand stands in contrast to Park et al. (2021) and Vana and Lambrecht (2021), who both report large effects of reviews on demand. We argue that the structure of the market, the design of Airbnb's reputation system, and the capacity constraints of Airbnb listings are critical for understanding these findings. If sellers are expected to quickly accumulate reviews, then the effect of a marginal review is likely to be small. The effect of incentivized reviews is further reduced by the rounding of ratings to a coarse average. Both the speed at which sellers can accumulate reviews on Airbnb and the rounding of ratings in the platform's reputation system caused the incremental effects of one review on the platform to be small. Sellers on Airbnb also have limited capacity, putting a ceiling on the impact that any intervention can have on quantities sold. One might worry that, although Airbnb exhibits these qualities, other marketplaces may not. However, we also find evidence of these marketplace dynamics in a scraped data set that describes reviewing behavior on a large home improvement services platform²² and suspect that many other marketplaces have similar dynamics.

A key goal of a reputation system is to create good matches between buyers and sellers. In contrast, we find that incentivized reviews caused worse matches on the platform whether measured by subsequent ratings or guest return rates to the platform. We argue that these worse matches were caused at least in part by the fact that incentivized ratings were more inflated conditional on a rating level. In particular, conditional on a given star rating, incentivized ratings represented a worse experience than nonincentivized ratings. This finding highlights that it is important to look beyond the ratings distribution when determining whether a policy increases or decreases review inflation.

Our negative evaluation of a specific incentivized review program does not preclude other interventions targeted at increasing review rates from having positive effects. Our treatment induced reviews for a specific set of transactions and had imperfect compliance; only 37.05% of treated transactions were reviewed. A policy that induced reviews for a different subset of transactions, for example, those with customer complaints, could have different effects on market outcomes. Alternatively, a more intensive incentivized review policy that encourages reviews for many transactions per seller could have larger effects.

Finally, the incentivized review policy that we study is not well-suited toward solving the cold-start problem in online marketplaces. In order to solve the cold-start problem, a platform needs to consider alternative interventions, such as hiring new sellers directly as in Pallas (2014). Other policies that may solve the cold-start problem include subsidizing transactions with new sellers, boosting new sellers in search, and hiring "mystery shoppers" to examine the quality of new inventory. Whether these policies would be successful is an open question that we leave for future work.

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1. Delaying disclosure or publication until any intellectual property rights have been adequately protected.
2. Removing Airbnb confidential information from the work product.
3. Compliance with Airbnb's obligations as it relates to applicable laws, including the Data Protection Law, security laws, confidentiality requirements, and contractual commitments.

4. Adding an appropriate disclosure concerning Airbnb's support of the project.

Endnotes

- ¹ Each individual review helps other market participants but not the person who provides the review.
- ² As an example, Online Figure C.1 shows an email sent as part of an incentivized review program run by the eCommerce website Girlfriend Collective.
- ³ We were not able to calculate the exact cost of the experiment because data on coupon redemption was not available to us, but our best guess is that the policy cost on the order of \$250,000.
- ⁴ We document a similarly fast review arrival rate on a large online marketplace for home improvement services in Online Section B.8.
- ⁵ Other work that uses field experiments to study the effects of changes to reputation system design includes Cabral and Li (2015), Benson et al. (2020), Hui et al. (2023), Fradkin et al. (2021), Garg and Johari (2021). A preliminary analysis of this experiment focusing on the first month is presented in Fradkin et al. (2015).
- ⁶ A related literature studies "fake" reviews (Mayzlin et al. 2014, Luca and Zervas 2016, He et al. 2022); however, fake reviews are less of a concern in our research setting because all Airbnb reviews are linked to verified transactions.
- ⁷ In Online Appendix A, we develop a theoretical framework that simplifies the framework put forth in Acemoglu et al. (2022) and clarifies the conditions under which we expect incentivized reviews to increase demand and/or the utility of buyers.
- ⁸ Laouénan and Rathelot (2020) and Cui et al. (2020) study the effects of Airbnb reviews on market outcomes with a focus on discrimination.
- ⁹ More detailed information on the timing of review email dispatch during our experiment is found in Online Appendix B.1.
- ¹⁰ We describe our methodology for text classification and the details of our results about sentiment in Online Appendix B.2.
- ¹¹ In Online Appendix A, we use a theoretical model to explore the exact circumstances under which reviews induced by the incentives increase market outcomes.
- ¹² In Online Figure C.10, we display the effects of the incentivized review treatment (in percentage terms) on demand outcomes for time horizons other than 120 days. We see similar results across time horizons with the peak occurring at 120 days. The effects in percentage terms shrink as the horizon expands, which reflects the temporary effects of the treatment. In Online Appendix B, we show that the effect on reservations comes from the intensive margin and that the estimates remain similar when adding controls.
- ¹³ This analysis requires two assumptions. First, that the coupon email does not change the type of review submitted by those who would review regardless of the email (the always takers). Second, that the email did not dissuade anyone from reviewing (no defiers). Also note that, in the case with no covariates, the estimated local average treatment effect of a review simply scales the estimate in Equation (1) by one over 12.86 percentage points, the causal effect of the coupon email on the review rate.
- ¹⁴ We also show in Online Appendices B.6 and B.7 that our lack of overall treatment effect on demand is not obscuring treatment effect heterogeneity with respect to ex ante expectations of listing demand or the star ratings of the reviews induced by the incentive to review.
- ¹⁵ We can measure the degree of supply constraint by dividing the number of inquiries by the number of listings contacted by market during the time that the experiment was conducted. The average listing in our experiment was booked in a market in which the

tightness (31.6) was much higher than the tightness in a typical market (18.1).

- ¹⁶ Rounding to half a star is a common design online and is used by Amazon, Etsy, and Yelp.
- ¹⁷ Analysis of pricing is found in Online Appendix B.5.
- ¹⁸ User return rates to the platform are used as a measure of customer satisfaction in Nosko and Tadelis (2015) and Farronato et al. (2020).
- ¹⁹ Hui et al. (2021) propose another reason that subsequent trips to treatment listings may result in lower ratings on average: low ratings may be autocorrelated because of belief-updating dynamics that affect review rates. Online Table C.3 shows that these results are not substantively affected by the inclusion of additional covariates in our regression model.
- ²⁰ The specific dynamics that could cause an incentivized review program to lead to worse matches are formalized in Online Appendix A.
- ²¹ Although this is the most natural way to measure inflation in review ratings, it requires conditioning on the first review that a listing receives. Thus far, we have avoided conditioning on the incidence or star rating of the focal review. Although conditioning on reviews has intuitive appeal, this approach to analysis suffers from severe omitted variable bias. Keeping this in mind, the analysis that follows should be interpreted with caution.
- ²² Online Appendix B.8 provides more detail on how these data were collected and on the analysis we conducted using it.

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